SQL Exercise 4

Kutlaqo Icqone

(1)

(1) SELECT EmployeeID, FirstName, LastName,
Department,
Salary,
Project ID, ProjectName, Budget,
Status

FROM Employees AS A
INNER JOIN Projects AS B
ON A. EmployeeID = B. EmployeeID;

EmployeID	Firstname	Lastname	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	7000	101	AI Development	[illegible]	Completed
1	John	Doe	IT	7000	103	Cybersecurity	[illegible]	Pending
2	Alice	Smith	HR	6000	102	Employee T	50 000	Ongoing
3	Bob	Johnson	Finance	5000	104	Financial A	90 000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market E	60000	Completed
6	Michael	Clark	Finance	80000	106	Risk Man	80000	Pending

(2) SELECT EmployeeID, FirstName, LastName, Department, Salary,
ProjectID, ProjectName, Budget, Status

FROM Employees AS A
LEFT JOIN Projects AS B
ON A. Employees = B. Projects;

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	[illegible]	101	AI Development	[illegible]	Completed
1	John	Doe	IT	[illegible]	103	Cybersecurity Audit	[illegible]	Pending
2	Alice	Smith	HR	[illegible]	102	Employee Training	[illegible]	Ongoing
3	Bob	Johnson	Finance	[illegible]	104	Financial Analysis	[illegible]	Ongoing
4	David	Brown	IT	[illegible]	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	[illegible]	105	Marketing Expansion	[illegible]	Completed
6	Michael	Clark	Finance	[illegible]	106	Risk Management	[illegible]	Pending

(3) SELECT ProjectID, ProjectName, Budget, Status, EmployeeID,
FirstName, LastName, Department, Salary
FROM Employees AS A
RIGHT JOIN Projects AS B
ON A. EmployeeID = B. EmployeeID;

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	[illegible]	Completed	1	John	Doe	IT	90000
102	Employee Training	[illegible]	Ongoing	2	Alice	Smith	HR	65000
103	Cybersecurity Audit	[illegible]	Pending	1	John	Doe	IT	70000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	95000
105	Marketing Expansion	[illegible]	Completed	5	Emma	Wilson	Sales	60000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	88000

(4) SELECT EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status
FROM Employees AS A
FULL OUTER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Dev	100000	Completed
1	John	Doe	IT	70000	103	Cybersec	75000	Pending
2	Alice	Smith	HR	60000	102	Employee A	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Ex	60000	Completed
6	Michael	Clark	Finance	85000	106	Risk Management	55000	Pe

(5) SELECT City AS Location FROM Employees
UNION
SELECT Status AS Location. FROM Projects;

Location
New York
Los Angeles
Toronto
London
Sydney
Completed
Ongoing
Pending

(6) SELECT City As location FROM Employees
UNION ALL
SELECT Status AS Location FROM Projects;

Location
New York
Los Angels
Toronto
London
Sydney
New York
Completed
Pendin
Ongoing
Ongoing
Pending

(7) SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Salary > 70 000;

EmployeID	FirstName	LastNam	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80 000

(8) SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Department IN ('IT', 'Finance');

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70 000
3	Bob	Johnson	Finance	75 000
4	David	Brown	IT	72 000
6	Michael	Clark	Finance	80 000

(9) SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE status NOT 'Completed';

ProjectID	ProjectName	Budget	Status
102	Employee Training	80000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk management	80000	Pending

(10) SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Budget > 70000 AND Status <> 'Completed';

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk management	80000	Pending

(11) SELECT EmployeeID, FirstName, LastName, Department, Salary, City
FROM Employees
WHERE City IN ('New York', 'Toronto')
ORDER BY Salary DESC;

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

(12) SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

(13) SELECT Department,
SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department
ORDER BY TotalSalary DESC;

Department	Total Salary
Finance	155000
IT	142000
Sales	65000
HR	60000

(14)

```
SELECT City, AVG(Salary) AS Average_salary
FROM Employees
GROUP BY City
HAVING AVG(Salary) > 65000;
```

City	Average_Salary
New York	75000
Toronto	75000
London	72000

(15)

```
SELECT Department, COUNT(EmployeID)
AS Number_of_employees
FROM Employees
GROUP BY Department
HAVING COUNT(EmployeID) > 1;
```

Department	Number_of_employees
IT	2
Finance	2

(16)

```
SELECT Status, COUNT (Project ID)
AS Project Count
FROM Projects
GROUP BY Status
HAVING COUNT(Project ID) >=2;
```

Status	Project Count
Completed	2
Ongoing	2
Pending	2

(17)

```
SELECT EmployeeID, FirstName, LastName,
SUM( Budget ) AS Total Project Budget
FROM Employees AS A
JOIN Projects AS B
ON A. EmployeeID = B EmployeeID
GROUP BY EmployeeID, FirstName, LastName
HAVING SUM( Budget ) > 150 000;
```

EmployeeID	FirstName	LastName	Total Project Budget
1	John	Doe	175 000